Linear Algebra Recap

Quantum Physics, 1st Exercise Sheet – Solutions

Exercise 1: Scalar and Dyadic Products

Given:

$$\mathbf{v} = \begin{pmatrix} 1 \\ i \\ 2 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 2 \\ 1 \\ i \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 5 \\ 2i \\ 1+i \end{pmatrix}$$

Tasks:

a) Compute all possible scalar products and norms.

Solution. $\langle v|v\rangle = 6$, $\langle v|w\rangle = 2+i$, $\langle v|u\rangle = 9+2i$, $\langle w|w\rangle = 6$, $\langle w|u\rangle = 11+i$, $\langle u|u\rangle = 31$ □

b) Compute the dyadic (outer) product $\mathbf{v} \circ \mathbf{w} = |v\rangle \langle w|$.

Solution.

$$|v\rangle \langle w| = \begin{pmatrix} 2 & 1 & -i \\ 2i & i & 1 \\ 4 & 2 & -2i \end{pmatrix}$$

□

c) Write the dyadic product in braket notation using the standard basis vectors:
$|0\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, |1\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, |2\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

Solution. $|v\rangle \langle w| = 2\,|0\rangle \langle 0| + |0\rangle \langle 1| - i\,|0\rangle \langle 2| + 2i\,|1\rangle \langle 0| + i\,|1\rangle \langle 1| + |1\rangle \langle 2| + 4\,|2\rangle \langle 0| + 2\,|2\rangle \langle 1| - 2i\,|2\rangle \langle 2|$ □

Exercise 2: Projector of a Single Vector

Given:

$$\mathbf{u} = \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix}$$

Tasks:

a) Construct the projector $P = \frac{|\mathbf{u}\rangle\langle\mathbf{u}|}{\langle\mathbf{u}|\mathbf{u}\rangle}$.

Solution. $\langle\mathbf{u}|\mathbf{u}\rangle = 2$

$$P = \frac{1}{2} \begin{pmatrix} 1 & -i & 0 \\ i & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

□

b) Compute $P\mathbf{u}$.

Solution.

$$P\mathbf{u} = \frac{1}{2} \begin{pmatrix} 1 & -i & 0 \\ i & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} = \mathbf{u}$$

□

c) Determine the rank (dimension) of P.

Solution. Calculate the linearly independent rows: the second row is i times the first row, while the third row is 0 times the first row, so the there is only one linearly independent row, which means that the rank of P is 1. □

d) Prove that P is a projection.

Solution. P is a projection if it is self-adjoint and $P^2 = P$. Its transpose conjugate is itself, therefore it is indeed self-adjoint. Moreover

$$P^2 = \frac{1}{4}\begin{pmatrix}1 & -i & 0\\ i & 1 & 0\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix}1 & -i & 0\\ i & 1 & 0\\ 0 & 0 & 0\end{pmatrix} = \frac{1}{4}\begin{pmatrix}2 & -2i & 0\\ 2i & 2 & 0\\ 0 & 0 & 0\end{pmatrix} = P$$

□

Exercise 3: Eigenvalues and Eigenvectors of a 2x2 Hermitian Matrix

Given:

$$H = \begin{pmatrix}5 & 1-2i\\ 1+2i & 3\end{pmatrix}$$

Tasks:

a) Compute the characteristic polynomial.

Solution. $(5-\lambda)(3-\lambda) - 5 = \lambda^2 - 8\lambda + 10 = 0$ □

b) Find the eigenvalues.

Solution. $\lambda_{1,2} = \frac{8 \pm \sqrt{64-40}}{2} = 4 \pm \sqrt{6}$ □

c) Find normalized eigenvectors. $H\,|\tilde{v}_1\rangle = \lambda_1\,|\tilde{v}_1\rangle$

Solution.

$$\begin{pmatrix}5 & 1-2i\\ 1+2i & 3\end{pmatrix}\begin{pmatrix}1\\ \alpha_1\end{pmatrix} = (4+\sqrt{6})\begin{pmatrix}1\\ \alpha_1\end{pmatrix} \iff 5+(1-2i)\alpha = 4+\sqrt{6} \iff \alpha = \frac{\sqrt{6}-1}{1-2i} = \frac{(\sqrt{6}-1)(1+2i)}{5}$$

$$|v_1\rangle = |\tilde{v}_1\rangle / |\tilde{v}_1| = \sqrt{\frac{5}{12-2\sqrt{6}}}\begin{pmatrix}1\\ \frac{(\sqrt{6}-1)(1+2i)}{5}\end{pmatrix} = \frac{1}{\sqrt{60-10\sqrt{6}}}\begin{pmatrix}5\\ (\sqrt{6}-1)(1+2i)\end{pmatrix}$$

$$\begin{pmatrix}5 & 1-2i\\ 1+2i & 3\end{pmatrix}\begin{pmatrix}1\\ \alpha_2\end{pmatrix} = (4-\sqrt{6})\begin{pmatrix}1\\ \alpha_1\end{pmatrix} \iff 5+(1-2i)\alpha = 4-\sqrt{6} \iff \alpha = -\frac{\sqrt{6}+1}{1-2i} = -\frac{(\sqrt{6}+1)(1+2i)}{5}$$

$$|v_2\rangle = |\tilde{v}_2\rangle / |\tilde{v}_2| = \sqrt{\frac{5}{12+2\sqrt{6}}}\begin{pmatrix}-1\\ \frac{(\sqrt{6}+1)(1+2i)}{5}\end{pmatrix} = \frac{1}{\sqrt{60+10\sqrt{6}}}\begin{pmatrix}-5\\ (\sqrt{6}+1)(1+2i)\end{pmatrix}$$

□

d) Verify orthonormality of eigenvectors.

Solution.

$$\begin{pmatrix}5 & (\sqrt{6}-1)(1-2i)\end{pmatrix}\begin{pmatrix}-5\\ (\sqrt{6}+1)(1+2i)\end{pmatrix} = -25 + (\sqrt{6}-1)(\sqrt{6}+1)(1-2i)(1+2i) = 0$$

□

e) Write the spectral decomposition $H = \lambda_1|v_1\rangle\langle v_1| + \lambda_2|v_2\rangle\langle v_2|$.

Solution.

$$H = \frac{4+\sqrt{6}}{12-2\sqrt{6}}\begin{pmatrix}5 & (\sqrt{6}-1)(1-2i)\\ (\sqrt{6}-1)(1+2i) & (7-2\sqrt{6})\end{pmatrix} + \frac{4-\sqrt{6}}{12+2\sqrt{6}}\begin{pmatrix}5 & -(\sqrt{6}+1)(1-2i)\\ -(\sqrt{6}+1)(1+2i) & (7+2\sqrt{6})\end{pmatrix}$$

□

f) *Optional:* Prove that if an $n \times n$ matrix has a spectral decomposition, than the matrix is hermitian if and only if all of its eigenvalues are real.

Solution. $H = \sum_j \lambda_j\,|j\rangle\,\langle j|$, if Hermitian:

$$\sum_j \lambda_j\,|j\rangle\,\langle j| = H = H^\dagger = \sum_j \lambda_j^*(|j\rangle\,\langle j|)^\dagger = \sum_j \lambda_j^*\,|j\rangle\,\langle j| \iff \lambda = \lambda^*,\ \forall j$$

□

Exercise 4: Matrix exponential

For an $n \times n$ Hermitian matrix, with eigendecomposition $H = \sum_i \lambda_i |v_i\rangle \langle v_i|$, one can define any analytical function f of the matrix as

$$f(A) = \sum_i f(\lambda_i) |v_i\rangle \langle v_i| .$$

Given:

$$H = \begin{pmatrix} 5 & 1-2i \\ 1+2i & 3 \end{pmatrix}$$

Tasks:

a) Using spectral decomposition $H = \lambda_+ |v_+\rangle\langle v_+| + \lambda_- |v_-\rangle\langle v_-|$, compute $U(t) = e^{-iH}$.

Solution.

$$e^{-iH} = \frac{e^{-i(4+\sqrt{6})}}{12-2\sqrt{6}} \begin{pmatrix} 5 & (\sqrt{6}-1)(1-2i) \\ (\sqrt{6}-1)(1+2i) & (7-2\sqrt{6}) \end{pmatrix} + \frac{e^{-i(4-\sqrt{6})}}{12+2\sqrt{6}} \begin{pmatrix} 5 & -(\sqrt{6}+1)(1-2i) \\ -(\sqrt{6}+1)(1+2i) & (7+2\sqrt{6}) \end{pmatrix}$$

Moreover:

$$\frac{5}{6 \pm \sqrt{6}} = 1 - \frac{1 \pm \sqrt{6}}{6 \pm \sqrt{6}} = 1 \mp \frac{1}{\sqrt{6}}, \; \frac{(\sqrt{6} \pm 1)(1 \pm 2i)}{6 \pm \sqrt{6}} = \frac{1 \pm 2i}{\sqrt{6}}, \; \frac{7 \pm 2\sqrt{6}}{6 \pm \sqrt{6}} = 1 + \frac{1 \pm \sqrt{6}}{6 \pm \sqrt{6}} = 1 \pm \frac{1}{\sqrt{6}}$$

Thus:

$$e^{-iH} = \frac{e^{-i(4+\sqrt{6})}}{2} \begin{pmatrix} 1+\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}}(1-2i) \\ \frac{1}{\sqrt{6}}(1+2i) & 1-\frac{1}{\sqrt{6}} \end{pmatrix} + \frac{e^{-i(4-\sqrt{6})}}{2} \begin{pmatrix} 1-\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}}(1-2i) \\ -\frac{1}{\sqrt{6}}(1+2i) & 1+\frac{1}{\sqrt{6}} \end{pmatrix}$$

□

b) Verify $U = e^{-iH}$ is unitary.

Solution.

$$U^\dagger = \frac{e^{i(4+\sqrt{6})}}{2} \begin{pmatrix} 1+\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}}(1-2i) \\ \frac{1}{\sqrt{6}}(1+2i) & 1-\frac{1}{\sqrt{6}} \end{pmatrix} + \frac{e^{i(4-\sqrt{6})}}{2} \begin{pmatrix} 1-\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}}(1-2i) \\ -\frac{1}{\sqrt{6}}(1+2i) & 1+\frac{1}{\sqrt{6}} \end{pmatrix}$$

$$U^\dagger U = \frac{1}{4} \begin{pmatrix} 1+\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}}(1-2i) \\ \frac{1}{\sqrt{6}}(1+2i) & 1-\frac{1}{\sqrt{6}} \end{pmatrix}^2 + \frac{1}{4} \begin{pmatrix} 1-\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}}(1-2i) \\ -\frac{1}{\sqrt{6}}(1+2i) & 1+\frac{1}{\sqrt{6}} \end{pmatrix}^2$$

$$\begin{pmatrix} 1+\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}}(1-2i) \\ \frac{1}{\sqrt{6}}(1+2i) & 1-\frac{1}{\sqrt{6}} \end{pmatrix}^2 = \begin{pmatrix} 2+\frac{2}{\sqrt{6}} & \frac{2}{\sqrt{6}}(1-2i) \\ \frac{2}{\sqrt{6}}(1+2i) & 2-\frac{2}{\sqrt{6}} \end{pmatrix},$$

$$\begin{pmatrix} 1-\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}}(1-2i) \\ -\frac{1}{\sqrt{6}}(1+2i) & 1+\frac{1}{\sqrt{6}} \end{pmatrix}^2 = \begin{pmatrix} 2-\frac{2}{\sqrt{6}} & -\frac{2}{\sqrt{6}}(1-2i) \\ -\frac{2}{\sqrt{6}}(1+2i) & 2+\frac{2}{\sqrt{6}} \end{pmatrix}$$

These imply: $U^\dagger U = I$. □

Exercise 5: Pauli Matrix Decomposition and Eigenvalues

Theorem: The 2×2 Hermitian matrices form a four dimensional, real vector space, and the Pauli matrices: $\sigma_0, \sigma_x, \sigma_y, \sigma_z$ form a basis in that vector space.

Practical meaning: any 2×2 Hermitian matrix can be written as $H = a_0\sigma_0 + a_x\sigma_x + a_y\sigma_y + a_z\sigma_z$, where a_0, a_x, a_y and a_z are real coefficients.

Proposition: The eigenvalues of a 2×2 Hermitian matrix are $\lambda_\pm = a_0 \pm |\vec{a}|$.

Given:

$$H = \begin{pmatrix} 5 & 1-2i \\ 1+2i & 3 \end{pmatrix}$$

Tasks:

a) Express $H = a_0\sigma_0 + a_x\sigma_x + a_y\sigma_y + a_z\sigma_z$.

Solution. $a_0 = \mathrm{Tr}H/2 = 4$, $a_x = \mathrm{Re}H_{01} = 1$, $a_y = -\mathrm{Im}H_{01} = 2$, $a_z = H_{00} - a_0 = 1$. □

b) Using $H = a_0\sigma_0 + \vec{a}\cdot\vec{\sigma}$, compute eigenvalues $\lambda_\pm = a_0 \pm |\vec{a}|$.

Solution. $|\vec{a}| = \sqrt{6} \implies \lambda_\pm = 4 \pm \sqrt{6}$. □

c) Find normalized eigenvectors using the direction of $\vec{a}$.

Solution. If $\vec{a} = |\vec{a}|(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$, then $|v_1\rangle = (\cos(\theta/2), e^{i\phi}\sin(\theta/2))^T$ and $|v_1\rangle = (-\sin(\theta/2), e^{i\phi}\cos(\theta/2))^T$. So we have to find θ and ϕ: $\theta = \arctan(\sqrt{5})$, $\phi = \arctan 2$. □

d) Using Pauli decomposition $H = a_0\sigma_0 + \vec{a}\cdot\vec{\sigma}$, compute $U = e^{-iH} = e^{-ia_0}[\cos(|\vec{a}|)\sigma_0 - i\sin(|\vec{a}|)\hat{a}\cdot\vec{\sigma}]$.

Solution. $a_0 = 4$, $\mathbf{a} = (1,2,1)$, $|\mathbf{a}| = \sqrt{6}$:

$$e^{-iH} = e^{-4i}\left(\cos(\sqrt{6})I - i\sin(\sqrt{6})\frac{\sigma_x + 2\sigma_y + \sigma_z}{\sqrt{6}}\right)$$

□

e) Compare the result of Exc. 5d) with the result in Exc. 4.

Solution.

$$e^{-4i}\left(\cos(\sqrt{6})I - i\sin(\sqrt{6})\frac{\sigma_x + 2\sigma_y + \sigma_z}{\sqrt{6}}\right) = e^{-4i}\begin{pmatrix}\cos(\sqrt{6}) - \frac{i\sin\sqrt{6}}{\sqrt{6}} & -\frac{i\sin\sqrt{6}}{\sqrt{6}}(1-2i) \\ -\frac{i\sin\sqrt{6}}{\sqrt{6}}(1+2i) & \cos(\sqrt{6}) + \frac{i\sin\sqrt{6}}{\sqrt{6}}\end{pmatrix}$$

The phase factor matches, so lets drop it and focus on the matrix only.

$$\begin{pmatrix}\cos(\sqrt{6}) - \frac{i\sin\sqrt{6}}{\sqrt{6}} & -\frac{i\sin\sqrt{6}}{\sqrt{6}}(1-2i) \\ -\frac{i\sin\sqrt{6}}{\sqrt{6}}(1+2i) & \cos(\sqrt{6}) + \frac{i\sin\sqrt{6}}{\sqrt{6}}\end{pmatrix} = \frac{e^{i\sqrt{6}}}{2}\begin{pmatrix}1 - \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}}(1-2i) \\ -\frac{1}{\sqrt{6}}(1+2i) & 1 + \frac{1}{\sqrt{6}}\end{pmatrix} + \frac{e^{-i\sqrt{6}}}{2}\begin{pmatrix}1 + \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}}(1-2i) \\ \frac{1}{\sqrt{6}}(1+2i) & 1 - \frac{1}{\sqrt{6}}\end{pmatrix}$$

□

Exercise 6: Projector of Two Non-Orthonormal Vectors (Optional)

Given:

$$\mathbf{v}_1 = \begin{pmatrix}1\\1\\0\end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix}1\\0\\1\end{pmatrix}$$

Tasks:

a) Construct the projector onto the subspace spanned by $\mathbf{v}_1$ and $\mathbf{v}_2$ using Gram-Schmidt orthonormalization.

Solution.

$$\mathbf{e}_1 = \mathbf{v}_1/|\mathbf{v}_1| = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\\0\end{pmatrix}$$

$$\tilde{\mathbf{v}}_2 = \mathbf{v}_2 - \langle\mathbf{e}_1|\mathbf{v}_2\rangle\mathbf{e}_1 = \mathbf{v}_2 - \frac{1}{\sqrt{2}}\mathbf{e}_1 = \frac{1}{2}\begin{pmatrix}1\\-1\\2\end{pmatrix}, \mathbf{e}_2 = \tilde{\mathbf{v}}_2/|\tilde{\mathbf{v}}_2| = \frac{1}{\sqrt{6}}\begin{pmatrix}1\\-1\\2\end{pmatrix}$$

$$P = |\mathbf{e}_1\rangle\langle\mathbf{e}_1| + |\mathbf{e}_2\rangle\langle\mathbf{e}_2| = \frac{1}{6}\begin{pmatrix}4 & 2 & 2\\ 2 & 4 & -2\\ 2 & -2 & 4\end{pmatrix},$$

$$P^2 = \frac{1}{36}\begin{pmatrix}4 & 2 & 2\\ 2 & 4 & -2\\ 2 & -2 & 4\end{pmatrix}\begin{pmatrix}4 & 2 & 2\\ 2 & 4 & -2\\ 2 & -2 & 4\end{pmatrix} = \frac{1}{36}\begin{pmatrix}24 & 12 & 12\\ 12 & 24 & -12\\ 12 & -12 & 24\end{pmatrix} = P$$

□

b) Verify that $P\mathbf{v}_1$ and $P\mathbf{v}_2$ lie in the span of $\mathbf{v}_1$ and $\mathbf{v}_2$.

Solution.

$$P\mathbf{v}_1 = \frac{1}{6}\begin{pmatrix} 4 & 2 & 2 \\ 2 & 4 & -2 \\ 2 & -2 & 4 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} 6 \\ 6 \\ 0 \end{pmatrix} = \mathbf{v}_1, \quad P\mathbf{v}_2 = \frac{1}{6}\begin{pmatrix} 4 & 2 & 2 \\ 2 & 4 & -2 \\ 2 & -2 & 4 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} 6 \\ 0 \\ 6 \end{pmatrix} = \mathbf{v}_2$$

□

c) Determine the rank of P.

Solution. The first two rows are linearly independent, but the third row can be obtained if one subtract the second row from the first one, i.e. the rank of P is two. □

Exercise 7: Logarithm of a 2x2 Unitary Matrix (Optional)

Given:

$$U = \begin{pmatrix} \frac{-1+i}{2} & \frac{-1+i}{2} \\ -\frac{1+i}{2} & \frac{1+i}{2} \end{pmatrix}$$

Tasks:

a) Using eigenvalue decomposition, find $H = i\log(U)$.

b) Using Pauli decomposition $U = b_0\sigma_0 + \vec{b}\cdot\vec{\sigma} = e^{-ia_0}(\cos(|\vec{a}|) - i\sin(|\vec{a}|)\hat{a}\cdot\vec{\sigma})$, compute $H = a_0\sigma_0 + \vec{a}\cdot\vec{\sigma}$.

c) Compare the results from both methods.